

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

Duration: 45 Minutes

QUESTIONS: 2

Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I T. Niranjana Kumar (192473007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Niranjana Kumar

1. Smart Home Automation System

Introduction

A Smart Home Automation System automatically controls lighting, temperature, and security based on user behaviour and environmental conditions. It uses sensors such as motion detectors, temperature sensors, and cameras to make intelligent decisions.

Types of Intelligent Agents

1. Simple Reflex Agent

- Works only on the current input.
- Uses condition–action rules.

Example:

- If motion is detected → Turn on the light.
- If no motion → Turn off the light.

Advantages:

- Fast response
- Easy to implement

Disadvantages:

- Cannot remember previous events.
- Cannot learn user preferences.

2. Model-Based Agent

- Maintains an internal model (memory) of the environment.
- Uses both current and previous information.

Example:

- Remembers that the living room lights are already ON.
- Adjusts temperature based on current and previous readings.

Advantages:

- Handles partially observable environments.
- Makes better decisions than a simple reflex agent.

Disadvantages:

- Requires more memory and processing.

3. Goal-Based Agent

- Makes decisions to achieve a specific goal.

Example:

- Goal: Maintain room temperature at **24°C**.
- Turns the AC or heater ON/OFF until the goal is reached.

Advantages:

- Selects actions that achieve desired outcomes.
- More flexible than reflex agents.

Disadvantages:

- May take longer to decide.

4. Utility-Based Agent

- Chooses the action with the highest overall benefit (utility).
- Considers comfort, energy efficiency, and security.

Example:

- Reduces AC usage to save electricity while keeping the room comfortable.
- Locks doors and turns off unused lights when no one is home.

Advantages:

- Provides the best overall performance.
- Balances multiple objectives.

Disadvantages:

- More complex to design.

Comparison Table

Agent Type	Memory	Goal	Example
Simple Reflex	No	No	Turn on light when motion is detected
Model-Based	Yes	No	Remember room status
Goal-Based	Yes	Yes	Maintain desired temperature
Utility-Based	Yes	Yes	Maximize comfort and save energy

Conclusion

Using a Hybrid Model-Based and Utility-Based Agent provides the best smart home experience by ensuring user comfort, reducing energy consumption, and improving home security.

2. Robot in an Unknown Maze

Introduction

A robot is placed in an unknown maze and must find the exit without knowing the map. It explores the maze using search algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the lowest path cost first.

Working

1. Start from the initial position.
2. Store paths in a priority queue.
3. Expand the node with the smallest cost.
4. Continue until the goal is found.

Advantages

- Finds the least-cost path.
- Works with different movement costs.
- Complete and optimal.

Disadvantages

- Uses more memory.
- Can be slow for large mazes.

Complexity

- **Time:** $O(b^{(1 + \lceil C^*/\epsilon \rceil)})$
- **Space:** $O(b^{(1 + \lceil C^*/\epsilon \rceil)})$

Iterative Deepening Search (IDS)

Definition

IDS repeatedly performs Depth-First Search with increasing depth limits until the goal is found.

Working

1. Search with depth limit 0.
2. Increase limit to 1.
3. Continue increasing the depth limit until the goal is reached.

Advantages

- Low memory usage.
- Complete when the branching factor is finite.
- Suitable when the goal depth is unknown.

Disadvantages

- Repeats node expansions.
- Less efficient than UCS when path costs differ.

Complexity

- **Time:** $O(b^d)$
- **Space:** $O(bd)$

Comparison

Feature	UCS	IDS
Search Type	Cost-based	Depth-based
Optimal	Yes	Yes (equal costs)
Memory	High	Low
Speed	Slower	Faster for shallow goals
Suitable for Unknown Cost	Yes	No

Intelligent Agent

A **Goal-Based Agent** is suitable because:

- The goal is to reach the maze exit.
- The robot chooses actions that move it toward the goal.
- It updates decisions as it explores the environment.

Best Combination

Goal-Based Agent + Uniform Cost Search (UCS)

Justification

- The Goal-Based Agent focuses on reaching the exit.
- UCS finds the minimum-cost path.
- It guarantees an optimal solution even when movement costs vary.
- The robot can safely navigate the maze while minimizing total travel cost.

Conclusion

For an unknown maze, Uniform Cost Search with a Goal-Based Agent is the best solution because it ensures the robot reaches the exit using the least-cost path while making intelligent decisions throughout the search process.